PRACTICAL 5
ANALYZING DATA USING TABLE CALCULATIONS IN TABLEAU

Aim:- To analyze data using Tableau table calculations and understand how to perform running totals, percentage calculations, moving averages, cumulative metrics, year-over-year growth, ranking, window functions, addressing, partitioning, and dynamic titles and labels.

Objectives:

After completing this practical, students will be able to:

1. Apply Quick Table Calculations such as Running Total and Percent of Total.
2. Create Moving Averages and cumulative metrics.
3. Perform Year-over-Year Growth Analysis.
4. Use ranking functions to identify Top N and Bottom N records.
5. Implement window functions such as WINDOW_SUM and WINDOW_AVG.
6. Customize addressing and partitioning for table calculations.
7. Create dynamic titles and labels using Tableau calculations.

(a) APPLY QUICK TABLE CALCULATIONS

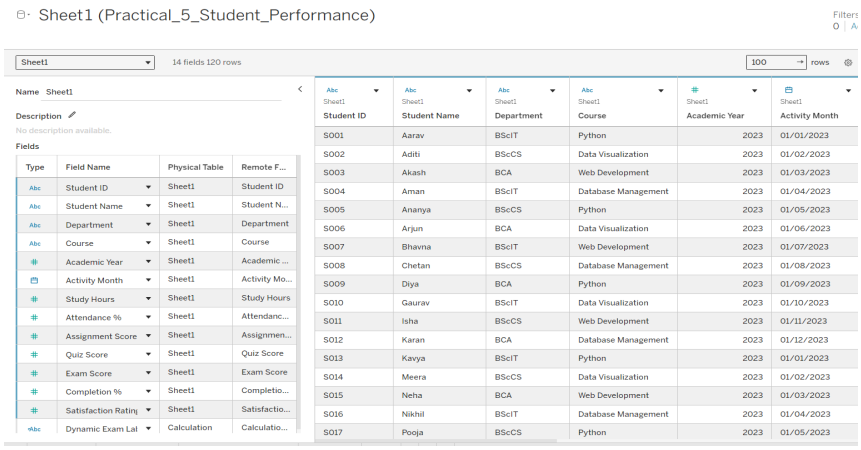
There are two parts:

- **(a1) Running Total**
- **(a2) Percent of Total**

(a1) Running Total of Study Hours

Aim: To calculate the cumulative Study Hours month by month using Tableau's Quick Table Calculation.

Excel Sheet:



Step 1:
Create a Worksheet
1. Open the Tableau workbook.
2. Click New Worksheet.
3. Rename the worksheet as:
Running Total

Step 2:
Add Activity Month
1. From the **Data Pane,** locate **Activity Month.**
2. Drag Activity **Month** to the **Columns** shelf.
3. If Tableau displays individual dates, click the drop-down arrow beside **Activity Month.**
4. Select **Month.**
The months will appear on the horizontal axis.

Step 3:
Add Study Hours
1. From the **Data Pane**, locate **Study Hours.**
2. Drag **Study Hours** to the **Rows** shelf.
3. Tableau will display:
SUM (Study Hours)

Step 4:
Apply Running Total
1. Right-click SUM(Study Hours) on the **Rows** shelf.
2. Select:
Quick Table Calculation → Running Total
Tableau will now calculate the cumulative Study Hours for each month.

Example:
Suppose the monthly Study Hours are:

Month	Monthly Study Hours	Running Total
January	405	405
February	441	846
March	322	1,168
April	320	1,488

For February:
Running Total = 405 + 441 = 846

Important Note About Labels

Sometimes the graph may display the original monthly value, such as 441, instead of the running total **846.**

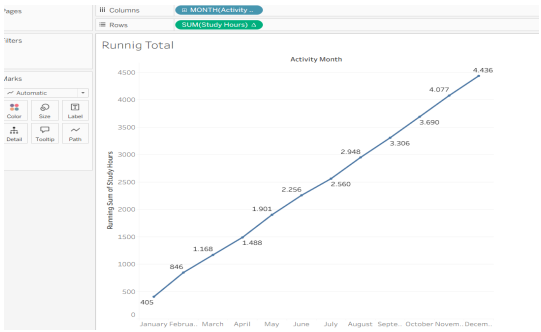
To correct this:

1. Go to the **Marks** card.
2. Click **Label.**
3. Remove the existing label field if required.
4. Drag SUM(Study Hours) from the Data Pane to **Label**.
5. Click the drop-down arrow on the SUM(Study Hours) pill on the Label shelf.
6. Select:

Quick Table Calculation → Running Total

Now both the graph and the labels will display the running-total values.

Output:



(a2) Percent of Total Exam Score

Aim: To calculate the percentage contribution of each month's Exam Score to the total Exam Score using Tableau's Quick Table Calculation.

Step 1:

Create a New Worksheet

1. Click **New Worksheet (+)**.
2. Rename the worksheet:

Percent of Total

Step 2:

Add Activity Month to Columns

1. Drag **Activity Month** to the **Columns** shelf.
2. Select **Month.**
3. If required, click the drop-down arrow on MONTH(Activity Month).
4. Select **Discrete.**

The columns should appear as:

Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec

Step 3:

Add Exam Score to Rows

1. Drag **Exam Score** to the **Rows** shelf.
2. Tableau will initially display:

SUM (Exam Score)

At this stage, Tableau is displaying the total Exam Score for each month.

Step 4:

Apply Percent of Total

1. Right-click the green SUM (Exam Score) pill on the **Rows** shelf.
2. Select:

Quick Table Calculation → Percent of Total

Tableau will convert the monthly Exam Scores into percentages of the overall total.

Step 5:

Display Percentage Values

1. Go to the **Marks** card.
2. Click **Label.**
3. Select **Show mark labels.**

The percentage values will now appear directly on the graph.

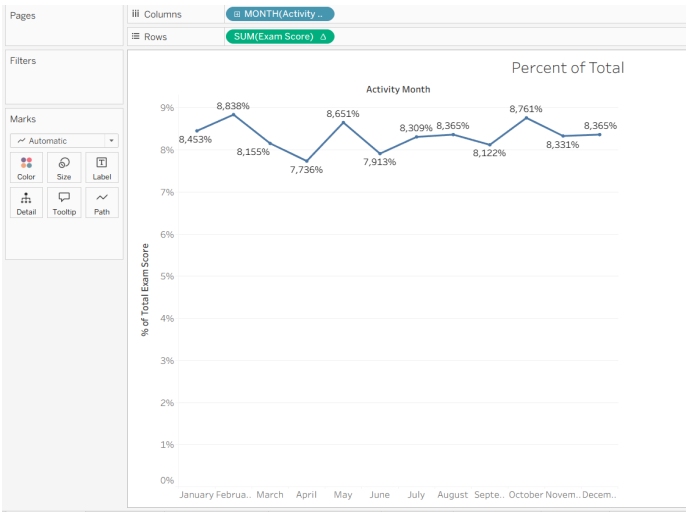
<u>Example Output:</u>

The graph may display values such as:

- January – 8.5%
- February – 8.8%
- March – 8.2%

The exact values depend on the dataset.

Output:



(b) MOVING AVERAGE AND CUMULATIVE METRICS
(b1) Three-Month Moving Average

Aim: To calculate a three-month moving average of Exam Score using Tableau.
A moving average helps identify the overall trend by averaging the current month with previous months.

Step 1:
Create a New Worksheet
1. Click **New Worksheet (+).**
2. Rename the worksheet:
Moving Average

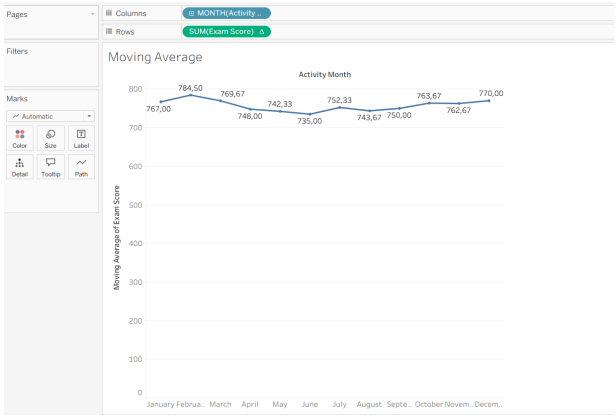
Step 2:
Add Activity Month
1. Drag **Activity Month** to **Columns.**
2. Select **Month.**
3. Make it **Discrete**, if necessary.
The columns should appear as:
Jan | Feb | Mar | Apr | May | Jun |...

Step 3:
Add Exam Score
1. Drag **Exam Score** to **Rows**.
2. Tableau will initially display the monthly Exam Score.

Step 4:
Apply Moving Average
1. Right-click SUM(Exam Score).
2. Select:
Quick Table Calculation → Moving Average
Tableau will now calculate a moving average.

Step 5:
Customize the Moving Average to Three Months
1. Right-click SUM (Exam Score) again.
2. Select **Edit Table Calculation**.
3. Choose **Moving Average**.
4. Set:
Previous values = 2
Next values = 0
This creates a three-month moving average because the calculation includes:
Previous 2 months + Current month

Output:



(b2) Cumulative Metric – Study Hours

Aim: To calculate cumulative Study Hours month by month.

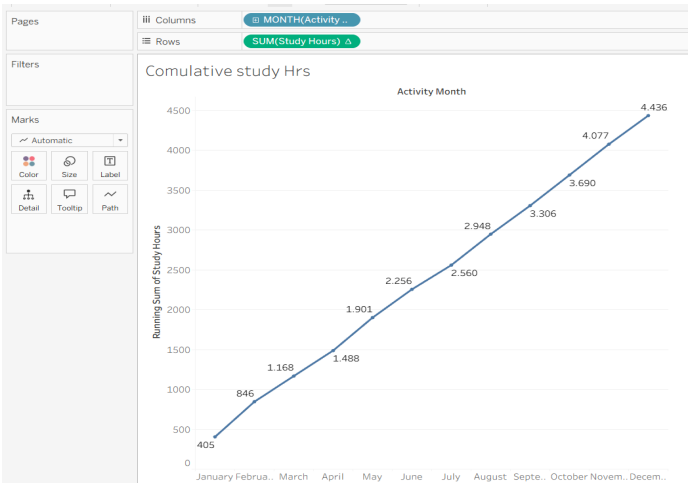
Step 1:
Create a New Worksheet
1. Create another worksheet.
2. Rename it:
Cumulative Study Hours

Step 2:
Add Activity Month
Drag **Activity Month** to **Columns** and select **Month.**

Step 3:
Add Study Hours
Drag **Study Hours** to **Rows.**

Step 4:
Apply Running Total
1. Right-click SUM(Study Hours).
2. Select:
Quick Table Calculation → Running Total
The resulting values represent cumulative Study Hours over time.
This calculation is similar to the Running Total calculation in part (a1), but here it is interpreted specifically as a cumulative metric.

Output:



(C) YEAR-OVER-YEAR GROWTH ANALYSIS

Aim: To compare the Exam Score of one academic year with the previous academic year and calculate the percentage growth.

The dataset contains the academic years:
2023, 2024, and 2025
Therefore, year-over-year growth can be calculated.

Step 1:
Create a New Worksheet
1. Click **New Worksheet (+).**
2. Rename the worksheet:
Year-over-Year Growth

Step 2:
Add Academic Year
1. From the Data Pane, locate **Academic Year**.
2. Drag it to **Columns**.
The columns should show:
2023 | 2024 | 2025

Step 3:
Add Exam Score
1. Drag **Exam Score** to **Rows.**
2. Tableau will display:
SUM (Exam Score)
The output will conceptually look like:

Academic Year	Exam Score
2023	Total for 2023
2024	Total for 2024
2025	Total for 2025

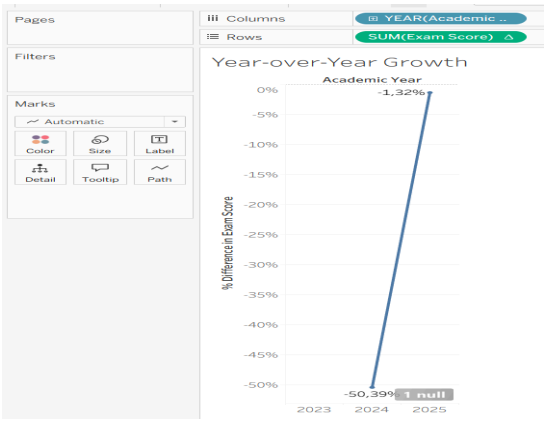
Step 4:
Apply Year-over-Year Growth
1. Right-click SUM (Exam Score).
2. Select:
Quick Table Calculation → Year-over-Year Growth
Tableau will calculate the percentage change compared with the previous year.

Example Output:

Academic Year	Exam Score	YoY Growth
2023	2,800	—
2024	2,950	5.36%
2025	3,100	5.08%

The first year is blank because there is no previous year for comparison.
Calculation for 2024
YoY Growth = ((2,950 − 2,800) / 2,800) × 100
= 5.36%
Calculation for 2025
YoY Growth = ((3,100 − 2,950) / 2,950) × 100
= 5.08%
Interpretation
A positive value such as **+5.36%** means that the Exam Score increased by 5.36% compared with the previous year.
A negative value such as **−5.36%** would mean that the Exam Score decreased by 5.36%.
Result
Year-over-Year growth in Exam Score was successfully calculated using Tableau table Calculations.

Output:



(d) RANKING FUNCTIONS – TOP N AND BOTTOM N

Two worksheets will be created:
1. Top 5 Students by Exam Score
2. Bottom 5 Students by Attendance
Because each student may appear in multiple records, AVG is used instead of SUM.
This provides a more meaningful student-level ranking.

(d1) Top 5 Students by Average Exam Score
Aim: To identify the five students with the highest average Exam Score.

Step 1:
Create a New Worksheet
Create a new worksheet and rename it:
Top 5 Students

Step 2:
Add Student Name
Drag **Student Name** to **Rows**.

Step 3:
Add Exam Score
Drag **Exam Score** to **Columns**.
Initially, Tableau will display:
SUM (Exam Score)

Step 4:
Change SUM to AVG
1. Click the drop-down arrow on SUM (Exam Score).
2. Select:

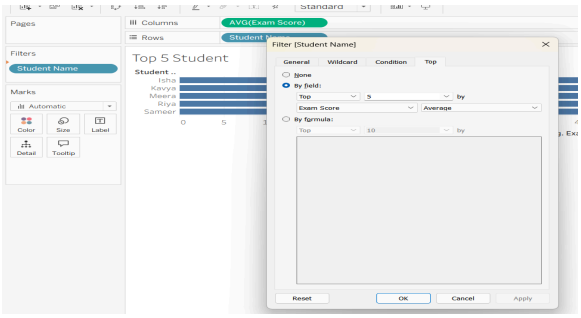
Measure → Average
The field will now become:
AVG (Exam Score)

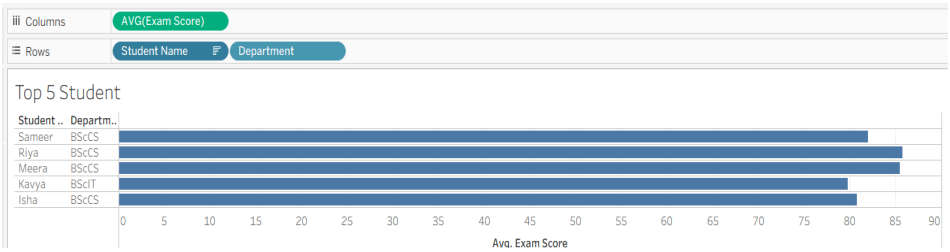
Step 5:
Create the Top 5 Filter
1. Drag **Student Name** to **Filters.**
2. In the Filter dialog box, select the **Top** tab.
3. Select **By field.**
4. Choose: **Top 5**
5. Select: **AVG (Exam Score)**
6. Click **OK**.

Step 6:
Sort the Results
Click the **Sort Descending** button to display the students from the highest average Exam Score to the lowest.

<u>Expected Result</u>
The worksheet will display the five students having the highest average Exam Score.

Output:





(d2) Bottom 5 Students by Average Attendance

Aim: To identify the five students with the lowest average Attendance Percentage.

Step 1:
Create a New Worksheet
Create a new worksheet and rename it:
Bottom 5 Students

Step 2:
Add Student Name
Drag **Student Name** to **Rows.**

Step 3:
Add Attendance %
Drag **Attendance %** to **Columns**.

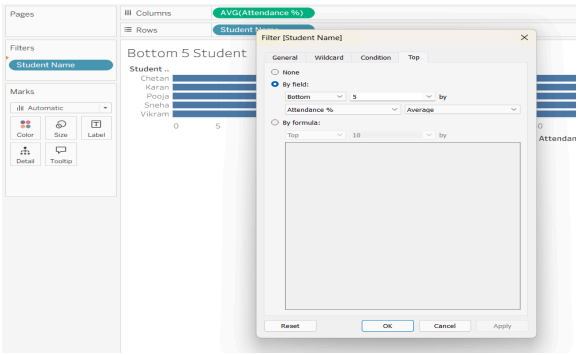
Step 4:
Change SUM to AVG
1. Click the drop-down arrow on SUM (Attendance %).
2. Select:
Measure → Average
The field will now become:
AVG (Attendance %)

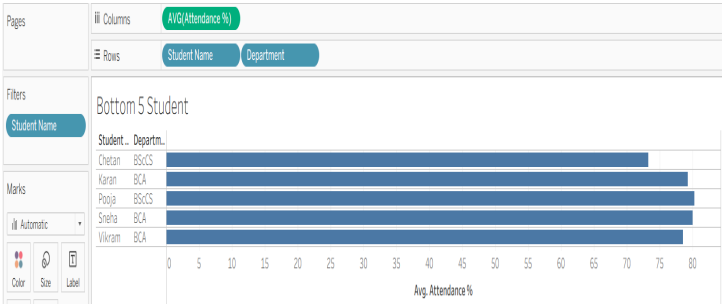
Step 5:
Apply Bottom 5 Filter
1. Drag **Student Name** to **Filters.**
2. Select the **Top** tab.
3. Choose **By field.**
4. Select:
Bottom 5
5. Select:
AVG(Attendance %)
6. Click **OK.**

Step 6:
Sort the Results
Click **Sort Ascending** to display the students with the lowest average Attendance first.

Expected Result
The worksheet will display the five students having the lowest average Attendance Percentage.

Output:





(e) IMPLEMENT WINDOW FUNCTIONS

Two window functions will be created:

1. WINDOW_SUM

2. WINDOW_AVG

(e1) WINDOW_SUM – Three-Month Study Hours

Aim: To calculate the total Study Hours within a three-month window.

Step 1:

Create a New Worksheet

Create a new worksheet and rename it:

WINDOW_SUM

Step 2:

Create a Calculated Field

1. Go to:

Analysis → Create Calculated Field

2. Name the calculated field:

- **3 Month Study Hours**

3. Enter the following formula:

WINDOW_SUM (SUM ([Study Hours]), -2, 0)

4. Click **OK**.
Explanation of the Formula

The formula:
WINDOW_SUM (SUM ([Study Hours]), -2, 0)

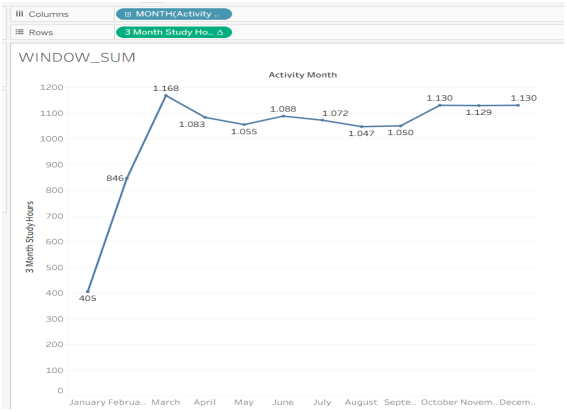
Means:

- -2 = include two previous values.
- 0 = include the current value.

Therefore, the calculation includes:
Previous 2 months + Current month
Step 3:
Add Activity Month
1. Drag Activity **Month** to **Columns**.
2. Select **Month.**

Step 4:
Add the Calculated Field
Drag **3 Month Study Hours** to **Rows.**

Output:



(e2) WINDOW_AVG – Three-Month Average Exam Score

Aim: To calculate the average Exam Score within a three-month window.

Step 1:
Create a Calculated Field
1. Go to:
Analysis → Create Calculated Field
2. Name the calculated field:

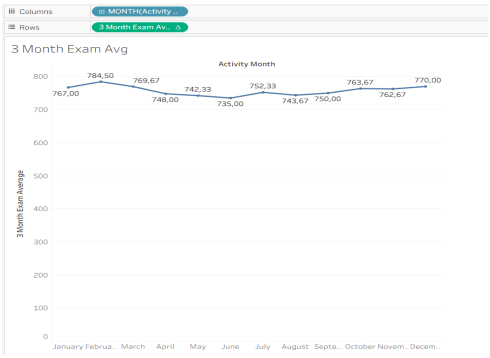
- **3 Month Exam Average**

3. Enter:
WINDOW_AVG (SUM([Exam Score]), -2, 0)
4. Click **OK.**

Step 2:
Add Activity Month
1. Drag **Activity Month** to **Columns.**
2. Select **Month.**

Step 3:
Add the Calculated Field
Drag **3 Month Exam Average** to **Rows.**
Tableau will calculate the average Exam Score for the current month and the previous two months.

Output:



(f) CUSTOMIZE ADDRESSING AND PARTITIONING

Aim: To understand how Tableau determines the direction of a table calculation and where the calculation should restart.
A three-month moving average will be calculated separately for each Course.

Addressing
Addressing tells Tableau the direction in which the table calculation should be performed.
In this practical:
Activity Month is the addressing dimension.
Therefore, Tableau calculates the values in the sequence:
January → February → March → April →…

Partitioning

Partitioning tells Tableau where the table calculation should restart.
In this practical:
Course is the partitioning dimension.
The calculation will be performed separately for each course, such as:

- Python
- Data Visualization
- Web Development
- Database Management

Steps to Customize Addressing and Partitioning

Step 1:
Create a New Worksheet
Create a new worksheet and rename it:
Addressing and Partitioning

Step 2:
Add Course
Drag **Course** to **Rows.**

Step 3:
Add Activity Month
Drag **Activity Month** to **Columns** and select **Month.**

Step 4:
Add Exam Score
Drag **Exam Score** to **Rows.**
Change the aggregation from:
SUM (Exam Score)
To:
AVG (Exam Score)

Step 5:
Apply Moving Average
Right-click the Exam Score field and select:
Quick Table Calculation → Moving Average

Step 6:
Edit the Table Calculation
1. Right-click the table-calculated field.
2. Select:

Edit Table Calculation

Step 7:
Select Specific Dimensions
Select:
Specific Dimensions

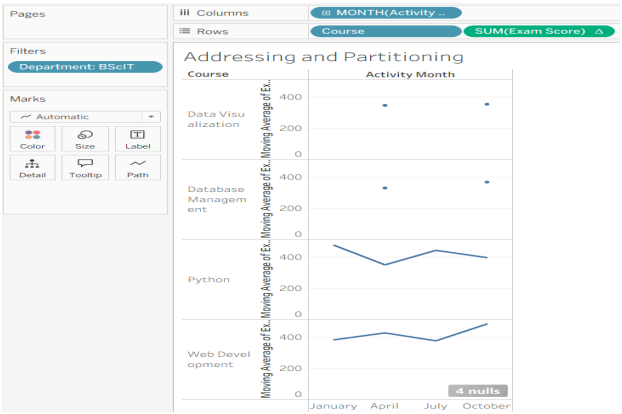
You will see:

- Course
- Activity Month

Make sure **Activity Month** is selected as the dimension through which Tableau addresses the calculation.
The **Course** dimension acts as the partition.

Result
Tableau will not calculate one moving average across all courses.
Instead, it will calculate a separate moving average for each course:
Python:
January → February → March → April →...
Data Visualization:
January → February → March → April →...
Web Development:
January → February → March → April →...
Database Management:
January → February → March → April →...
Expected Output
The worksheet will display separate lines or values for each course, with the moving average calculated independently for each course.

Output:



(g) CREATE DYNAMIC TITLES AND LABELS

This section demonstrates how Tableau can automatically change titles and labels based on the selected Course or other data values.

(g1) Dynamic Title

Aim: To create a worksheet title that changes dynamically according to the selected Course.

Step 1:
Create a New Worksheet
Create a new worksheet and rename it:
Dynamic Title

Step 2:
Add Course to Filters
1. Drag **Course** to **Filters.**
2. Select All.
3. Click **OK.**

Step 3:
Show the Course Filter
1. Right-click **Course** in the Filters area.
2. Select:
Show Filter
A Course filter will appear on the worksheet.

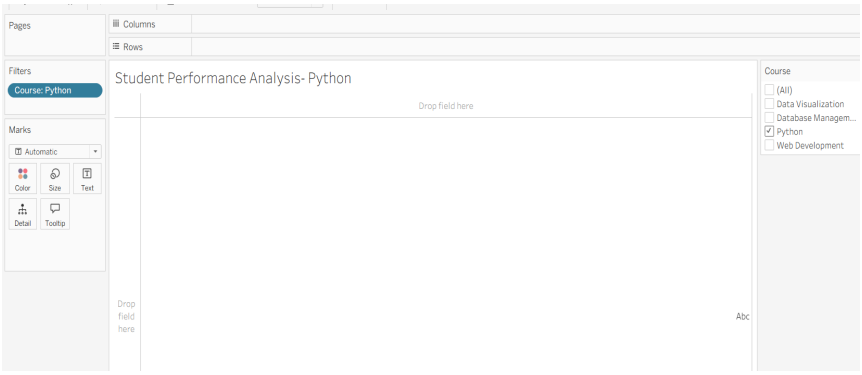
Step 4:
Show the Worksheet Title
Go to:
Worksheet → Show Title
The worksheet title will now be visible.

Step 5:
Edit the Title
1. Right-click the worksheet title.
2. Select **Edit Title**.
3. Enter:
Student Performance Analysis –
4. Select:
Insert → Course
The title will now contain the selected Course.

For example:
Student Performance Analysis – <Course>

Step 6:
Test the Dynamic Title
Selec**t Python** from the Course filter.
The title should display:
Student Performance Analysis – Python
Now select **Web Development.**
The title should automatically change to:
Student Performance Analysis – Web Development
This demonstrates that the worksheet title changes according to the selected filter Value.

Output:



(g2) Dynamic Label

Aim: To create a dynamic label that displays the Exam Score according to the selected Course or filter.

Step 1:
Create a Calculated Field
Go to:
Analysis → Create Calculated Field

Step 2:
Name the Calculated Field
Enter:
Dynamic Exam Label

Step 3:
Enter the Formula

Use the following formula:
"Exam Score:" + STR (ROUND (SUM ([Exam Score]), 0))
Click **OK**.

Explanation:
The formula performs the following operations:

- SUM ([Exam Score]) calculates the total Exam Score.
- ROUND (..., 0) rounds the result to a whole number.
- STR (...) converts the number into text.
- "Exam Score:" + adds descriptive text before the value.

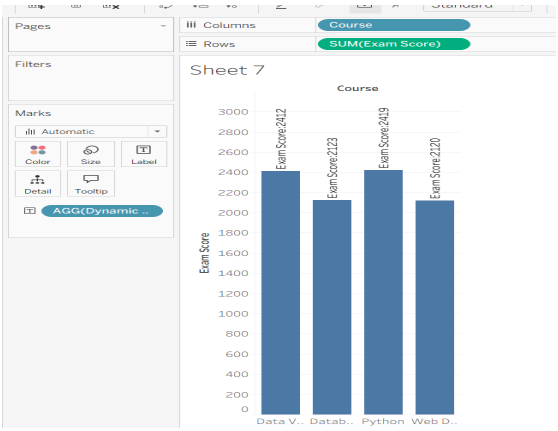
Step 4:
Create a Simple Chart
For example:

Drag **Course** to **Columns.**
Drag **Exam Score** to **Rows**.

Step 5:
Add the Dynamic Label
1. From the Data Pane, locate **Dynamic Exam Label**.
2. Drag it to **Label** on the **Marks** card.

Expected Output
The labels may display values such as:
Exam Score: 2,845
Or
Exam Score: 3,120
The exact values depend on the selected Course and the underlying data.
If a different Course is selected, the label automatically updates.

Output:



Conclusion:

Thus, the practical was successfully completed using Tableau table calculations. We learned to perform running totals, percentage of total, moving averages, cumulative metrics, year-over-year growth, ranking, window functions, addressing and partitioning. We also created dynamic titles and labels to make the data analysis more interactive and informative. These calculations helped us analyze student performance data effectively and understand trends and comparisons over time.